



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Software Construction And Development lab Report - 9

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PROGRAM: BSSE (5th Semester)

COURSE: Software Construction and Development

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DATE: 9/21/2025

DAY: Sunday

TASK 1: Add TODO Comments & Legal Comments to Car Rental System

Solution:

```
/**
 * CarRentalSystem.java
 *
 * LEGAL NOTICE:
 *
 * This program is created for educational purposes only.
 * Unauthorized copying, modification, or distribution is strictly prohibited.
 * The developer is not responsible for any misuse of the system.
 *
 * TODO COMMENTS ADDED:
 *
 * These comments mark areas where future improvements are required.
 */

import java.util.Scanner;

class Car {
    String strCarName;
    String strCarModel;
    String strCarNumber;

    void editCarDetails(String name, String model, String number) {

        // TODO: Validate the car name, must not be empty
        // TODO: Validate the car model format (e.g., year)

        strCarName = name;
        strCarModel = model;
        strCarNumber = number;
    }

    void printCarDetails() {
```

```
// TODO: Format output in table for clean display

System.out.println("Car Name: " + strCarName);
System.out.println("Car Model: " + strCarModel);
System.out.println("Car Number: " + strCarNumber);
}
}

class Customer {
    String strCustomerName;
    String strCustomerPhone;

    void editCustomerDetails(String name, String phone) {

        // TODO: Validate phone number length and digits only
        // TODO: Add Email Address field in next update

        strCustomerName = name;
        strCustomerPhone = phone;
    }

    void printCustomerDetails() {

        // TODO: Add CNIC field in future

        System.out.println("Customer Name: " + strCustomerName);
        System.out.println("Customer Phone: " + strCustomerPhone);
    }
}

class Rental {
    int intDays;
    double dblPricePerDay;
```

```

void editRentalDetails(int days, double price) {

    // TODO: Validate rental days (should not be negative)

    intDays = days;
    dblPricePerDay = price;
}

void printRentalDetails() {

    // TODO: Show breakdown: Base price + tax + discount

    System.out.println("Rental Days: " + intDays);
    System.out.println("Price per Day: " + dblPricePerDay);
    System.out.println("Total Price: " + (intDays * dblPricePerDay));
}
}

public class CarRentalSystem {

    // TODO: Add authentication (Admin Login)
    // TODO: Add database/file system integration

    public static void main(String[] args) {
        Scanner objScanner = new Scanner(System.in);

        Car objCar = new Car();
        Customer objCustomer = new Customer();
        Rental objRental = new Rental();

        System.out.print("Enter Car Name: ");
        String strName = objScanner.nextLine();
    }
}

```

```
        System.out.print("Enter Car Model: ");
        String strModel = objScanner.nextLine();

        System.out.print("Enter Car Number: ");
        String strNumber = objScanner.nextLine();

        objCar.editCarDetails(strName, strModel, strNumber);

        System.out.print("Enter Customer Name: ");
        String strCustName = objScanner.nextLine();

        System.out.print("Enter Customer Phone: ");
        String strCustPhone = objScanner.nextLine();

        objCustomer.editCustomerDetails(strCustName, strCustPhone);

        System.out.print("Enter Rental Days: ");
        int intDays = objScanner.nextInt();

        System.out.print("Enter Price per Day: ");
        double dblPrice = objScanner.nextDouble();

        objRental.editRentalDetails(intDays, dblPrice);

        System.out.println("\n--- Car Details ---");
        objCar.printCarDetails();

        System.out.println("\n--- Customer Details ---");
        objCustomer.printCustomerDetails();

        System.out.println("\n--- Rental Details ---");
        objRental.printRentalDetails();
    }
}
```

TASK 2: Improve Zakat Calculator with Explanation Comments

Solution:

```
/**
 * ZakatCalculator.java
 *
 * This program calculates Zakat based on Islamic rules.
 * Zakat is 2.5% of total wealth that has been held for one lunar year.
 *
 * IMPORTANT NOTES:
 * - Zakat applies only if wealth is above Nisab.
 * - Nisab amount changes every year depending on gold price.
 */

import java.util.Scanner;

public class ZakatCalculator {
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        // Zakat percentage (2.5% = 0.025)
        double zakat_rate = 0.025;

        double nisab = 135000;

        System.out.print("Enter your total amount: ");

        double total_amount = input.nextDouble();

        // Check if user's wealth qualifies for Zakat
        if (total_amount >= nisab) {

            // Zakat formula:
            // Zakat = total wealth × 2.5%
```

```

        double zakat_amount = total_amount * zakat_rate;

        System.out.println("Zakat Amount: " + zakat_amount);

    } else {

        // If wealth is below Nisab, no Zakat is due

        System.out.println("Your wealth is below the Nisab limit. No Zakat is required.");

    }

}
}
}

```

TASK 3: Example of Amplification Comments

Solution:

```

/**
 * AmplificationCommentsExample.java
 *
 * Amplification comments explain WHY a line of code is used.
 * They help in deeper understanding of system decisions.
 */

public class AmplificationCommentsExample {

    public static void main(String[] args) {

        int rentalDays = 10;

        int discount = 0;

        // AMPLIFICATION COMMENT:
        // We give a discount only if rental days exceed 7.
        // This encourages customers to book for longer durations.
        if (rentalDays > 7) {

            discount = 15; // 15% discount

```

```
}

// AMPLIFICATION COMMENT:
// StringBuilder is used instead of regular string concatenation
// because it is more efficient when building multiple strings dynamically.
StringBuilder message = new StringBuilder();
message.append("Total Rental Days = ").append(rentalDays);
message.append("\nDiscount Applied = ").append(discount).append("%");

System.out.println(message.toString());
}
}
```